

Calcium Competenet Cells Stock Generation

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

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WEDNESDAY, 12/07/2023

Started making calcium competent cell stocks according to protocol, starting with Day 3 of protocol.

- Inoculated 2 flasks of 50 mL with DH10-beta culture in the morning.
- Measured OD twice, stopped incubation at OD of 0.4 and 0.41.
- Volumes adjusted from protocol: 50 mL culture -> 30 mL MgCl₂ -> 40 mL CaCl₂ -> 10 mL CaCl₂ 85 nM, 15 % glycerol -> 400 uL CaCl₂ 85 nM, 15 % glycerol.

THURSDAY, 13/07/2023

Transformation to test efficiency (see protocol attached).

FRIDAY, 14/07/2023

Checked plates: negative control still showed growth on Amp plate.

Prepping gBlocks for Golden Gate cloning

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

Entry Created On: 13 Jul 2023 08:36:35 UTC

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THURSDAY, 13/07/2023

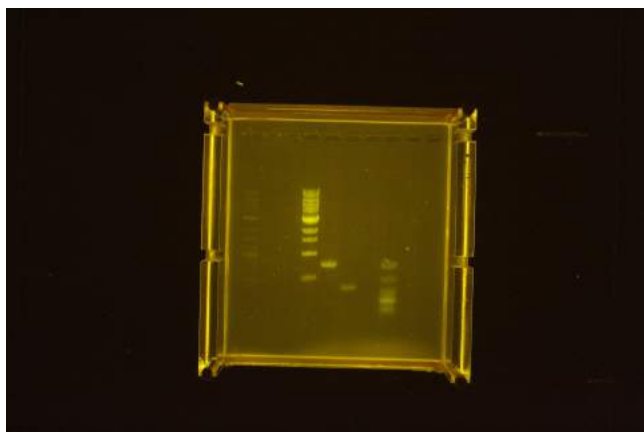
On Jul 13: received gBlocks for AcGFP sequence and HGFI hydrophobin with all mutated cysteines, according to instructions from IDT (linked to protocol). Vlad, Aby prepared gBlocks.

PCR ran with parameters from protocol. Brigitte helped with PCR.

FRIDAY, 14/07/2023

Gel for PCR products. 1% gel, ran for 45 min, 210 V. Vlad, Aby ran gel

 140723.JPG



Agarose gel 1%: PCR products

Lane 1, 3: 1kb ladder

Lane 2: pET28 plasmid

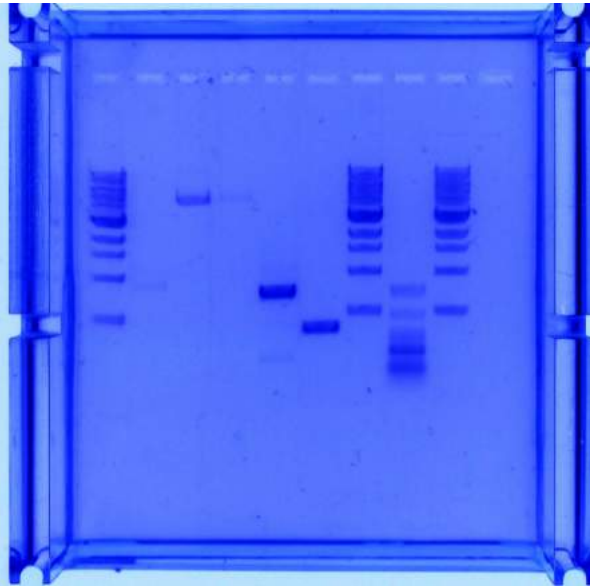
Lane 4: AcGFP fragment

Lane 5: HGFI_full_mutated fragment

Lane 7: Low MW DNA ladder

Gel after PCR purification:

IMG_0001_invert.jpg



Lane 1, 7,9: 1kb ladder

Lane 2: pET28 plasmid PCR amplified

Lane 3, 4: pET28 plasmid (circular)

Lane 5: AcGFP fragment

Lane 6: HGFI_full_mutated fragment

Lane 8: Low MW DNA ladder

Conclusion: PCR for plasmid backbone amplification did not work.

Golden Gate Cloning

Introduction

Protocol for the creation of Bsal restriciton sites and overhangs for Golden Gate Cloning.

Materials

- › Q5® High-Fidelity 2X Master Mix
- › Primers
- › DNA
- › Golden Gate Kit (Bsal, T4 ligase)

Procedure

Create Bsal sites by PCR

- ✓ 1. Use Q5 2x Master Mix to generate overhangs containing Bsal sites and complementarity sequences for ligation
- ✓ 2. Use NEB Primer annealing temperature calculator to determine PCR conditions (<https://tmcalculator.neb.com/#!/main>)

PCR Reaction Mix			
	A	B	C
1	Component	1x Volume (uL)	n x Volume (uL)
2	Q5 HF 2X Master mix	25	
3	Fwd primer (10 uM)	2.5	
4	Rev primer (10 uM)	2.5	
5	DNA template (10ng/uL)	1	
6	Water	19	



Cycling Parameters Short Insert



	A	B	C
1	STEP	TEMP	TIME
2	Initial Denaturation	98°C	30 seconds
3	30 Cycles	98°C	10 seconds
4		72°C	45 seconds
5			
6	Final Extension	72°C	2 minutes
7	Hold	16°C	

Cycling Parameters Plasmid



	A	B	C
1	STEP	TEMP	TIME
2	Initial Denaturation	98°C	30 seconds
3	30 Cycles	98°C	10 seconds
4		72°C	4 min
5			
6	Final Extension	72°C	2 minutes
7	Hold	16°C	

- ✓ 3. Dna input: <10 ng for gBlock and plasmid. Use 1 µL dilution of 10 ng/µL.
- ✓ 4. Incubate plasmid amplicon with 1 µL DpnI enzyme 37°C for 1h. Inactivation of enzyme is not required if next step is purification.
- ✓ 5. Purify PCR products using GeneJet PCR kit (Thermo Fischer)

Summary of Week July 11 - 14

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

Entry Created On: 13 Jul 2023 08:09:45 UTC

Entry Last Modified: 14 Jul 2023 14:28:37 UTC

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FRIDAY, 14/07/2023

This week we did:

- Train Brigitte, Aby on general techniques in the lab (replenish lab stocks, make competent cells, transformation)
- Received gBlocks for AcGFP and HGFI_full_mutated
- PCR-amplified pET28a, AcGFP and HGFI_full_mutated to introduce cutting sites for BsaI
- PCR for AcGFP and HGFI worked, see gel. pET28 amplicon did not work, will repeat PCR next week with multiple stocks from Pinheiro lab.

Amplify pET28a plasmid backbone

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

Entry Created On: 17 Jul 2023 07:52:47 UTC

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WEDNESDAY, 12/04/2017

MONDAY, 17/07/2023

Keywords (every experiment):

PCR, backbone plasmid

Background (every story):

PCR amplify pET28 plasmid backbone to introduce compatibility with Bsal-based Golden Gate cloning.

Key protocols and constructs (every experiment):

pET28a plasmid from Pinheiro lab, primers designed to create overhangs for Golden Gate.

Experimental details (every experiment):

Positive control: None

Negative control: Water

TUESDAY, 18/07/2023

Changed annealing temperature for PCR.

Set gradient centered around 66 ± 6 C.

Amplification parameters for gradient PCR:

Gradient PCR amplification			
	A	B	C
1	STEP	TEMP	TIME
2	Initial Denaturation	98°C	1 min
3	30 Cycles	98°C	15 s
4		60 ± 6	30 s
5		72°C	3 min 30 s
6	Final Extension	72°C	2 min
7	Hold	16°C	



WEDNESDAY, 19/07/2023

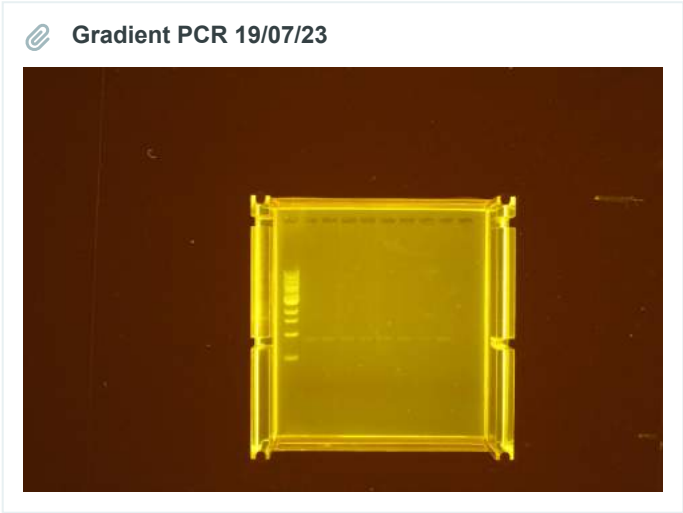
Run 1% agarose gel for gradient PCR.

Lane 1 : 1kb Plus ladder

Lane 2-9: Gradient PCR

Lane 10: Negative ctrl at 60C.

Gel image



Re-did PCR with KOD Hot Start polymerase.

Amplification of biosurfactant gBlocks

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

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MONDAY, 10/07/2023

Summary of Week July 17 - 21

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

Entry Created On: 22 Jul 2023 13:02:50 UTC

Entry Last Modified: 10 Oct 2023 15:25:30 UTC

Export Generated On: 10 Oct 2023 15:51:05 UTC

THURSDAY, 20/07/2023

1. Amplified sequences for hydrophobins and purified PCR products. Trained Tom on PCR, PCR purification and gel electrophoresis. PCRs for mutated hydrophobins failed, as primer pairs were mixed up. Needs repetition.
2. Tried optimizing amplification of pET28a backbone using separate routes: repeated with different stocks of plasmid, ran a gradient PCR for annealing temperature (66 ± 6 C), changed Q5 reaction mixtures from master mix to individual components, tried KOD Hot Start polymerase.
 - a. **Result: no amplification was possible, moved on with designs for pET29 plasmid**
3. Redesigned PCR primers for Golden Gate assemblies with pET29 plasmid from Sisi:
 - a. Designed primers that introduce SGGSGGS linker sequence upstream of sfGFP to create new plasmid backbone (pET29-linker-sfGFP).
 - b. Designed primers to introduce 8 surfactant sequences into plasmid backbone from point a) to create construct: pET29-Surfactant-SGGSGGS-sfGFP constructs.

Summary of Week of July 24 - 28: E.coli constructs v2 cloning

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

Entry Created On: 22 Jul 2023 14:02:41 UTC

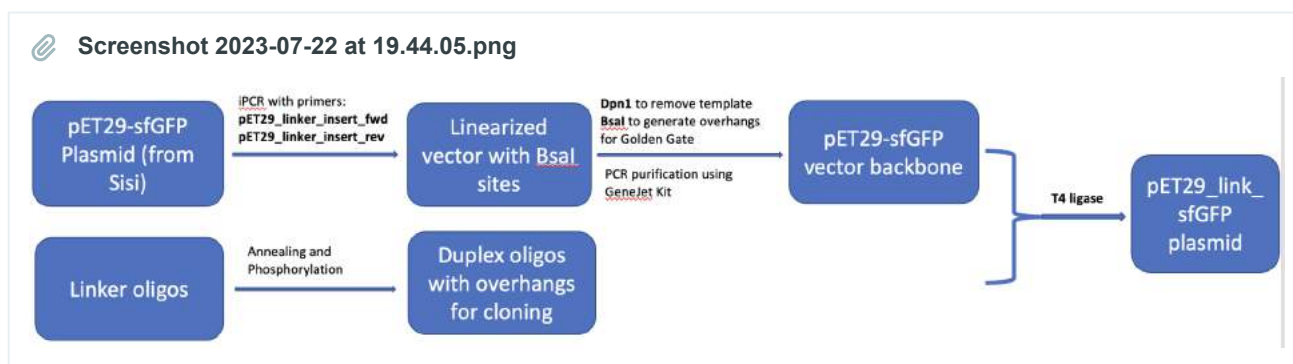
Entry Last Modified: 07 Oct 2023 11:36:41 UTC

Export Generated On: 10 Oct 2023 15:51:05 UTC

MONDAY, 24/07/2023

Create new constructs for E coli transformation.

Flowchart for intermediate plasmid assembly (plasmid with SGGSGGS linker upstream of sfGFP sequence).



[Protocol for prepping oligos for cloning](#)

In parallel, preparation of gBlocks for Golden Gate:

Summary of Monday, 24/07/2023

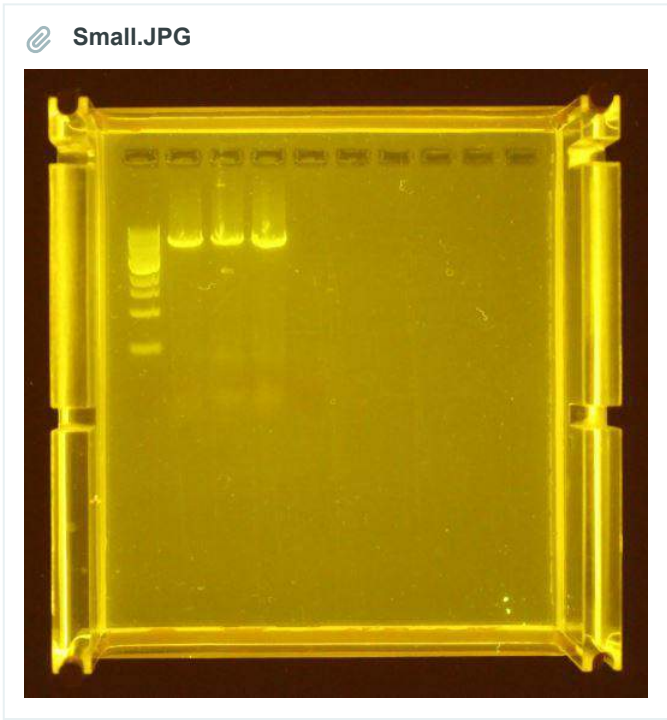
- Annealing temperature of the new primers have been created and is stored in [excel in Box](#)

TUESDAY, 25/07/2023

Primers and oligos arrived in the afternoon

Summary of Tuesday, 25/07/2023

- Diluting primers for PCR ready
 - All primers were diluted to 10 microM
- PCR amplification of plasmid (pET29-sfGFP from Sisi) using "Q5 plasmid 2" protocol (in Red Thermocycler)
 - 3 similar reactions were done
- Running gel for looking at result of PCR amplification
 - Result: bands were formed around 5.5 kb** (Lane 2-4 represent the three reactions done in three separate tubes)



- Digestion of the PCR amplification of plasmid (pET29-sfGFP from Sisi) using "Dig30-Inac20" protocol from Kiyo (in Red Thermocycler)
 - Digestions using enzymes DpnI (to remove parent plasmids) and BsaI (to create overhangs)
 - Materials used:

Table1		
	A	B
1	Materials	Volume (µL)
2	PCR product	45
3	DpnI	1
4	BsaI	1
5	CutSmart Buffer	10
6	MQ water	43
7		100



WEDNESDAY, 26/07/2023

Summary of Wednesday, 26/07/2023

- PCR purification using GenJet + Concentration determination
 - **Result:**

Table2



	A	B
1	Sample	Conc. (ng/μL)
2	1	24.61
3	2	55.92
4	3	73.36

- Annealing of linker oligos with "Annealing" protocol from Kiyo (in Black Thermocycler) + Concentration determination
 - **Result:**

Table3



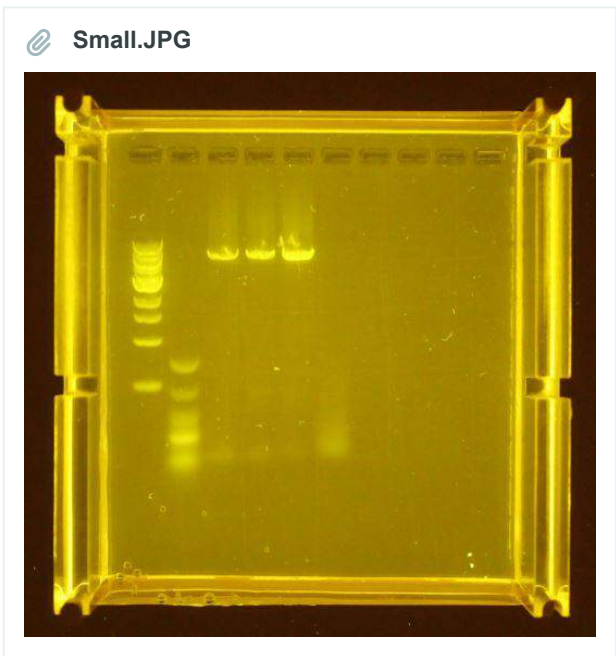
	A	B
1	Sample	Conc. (ng/μL)
2	Linker	6.22

- T4 ligation of linker (and phosphorylation of insert) with purified PCR result using "T4 Phos-Lig" protocol (in Red Thermocycler) to create pET29_link_sfGFP
 - Was done for 1:5 and 1:7 ratio (plasmid:linker)
- PCR amplification of plasmid (pET29-sfGFP from Sisi) using "Q5 plasmid 2" protocol (in Red Thermocycler)
 - The PCR was done again to create stock for vector backbone and also for practice

THURSDAY, 27/07/2023

Summary of Thursday, 27/07/2023

- Running gel for looking at result of PCR amplification
 - **Result:**



- PCR purification using GenJet + Concentration determination
 - **Result:**

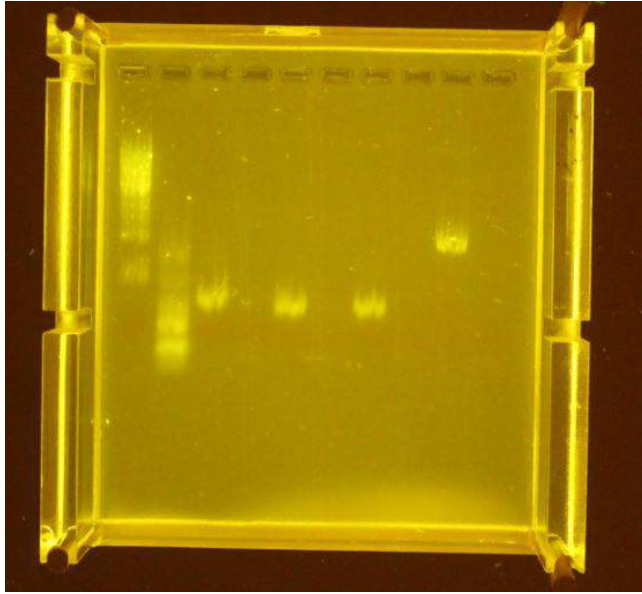
Table4			^
	A	B	
1	Sample	Conc. (ng/μL)	
2	1	54.71	
3	2	75.71	
4	3	62.9	

- Transformation of ligated plasmid (now pET29_link_sfGFP)
 - NEB 10-beta Competent *E. coli* cells
 - Protocol is as followed from [box](#)

FRIDAY, 28/07/2023

- PCR amplification of HFB1, HFB1 mutant, HFB2, and MBSP1.
 - Taq polymerase was used
 - Annealing temperature following in [excel in Box](#)
- Running gel for looking at result of PCR amplification
 - **Result: Sizes matched for HFB1, HFB1 mutant, HFB2, and MBSP1**

 Small.JPG



SUNDAY, 30/07/2023

- Overnight cultures in liquid media for sequencing
 - 3 colonies were picked from 1:5 ratio ligation of linker

Planning of Week July 31 - Aug 4

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

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Entry Last Modified: 07 Oct 2023 11:47:13 UTC

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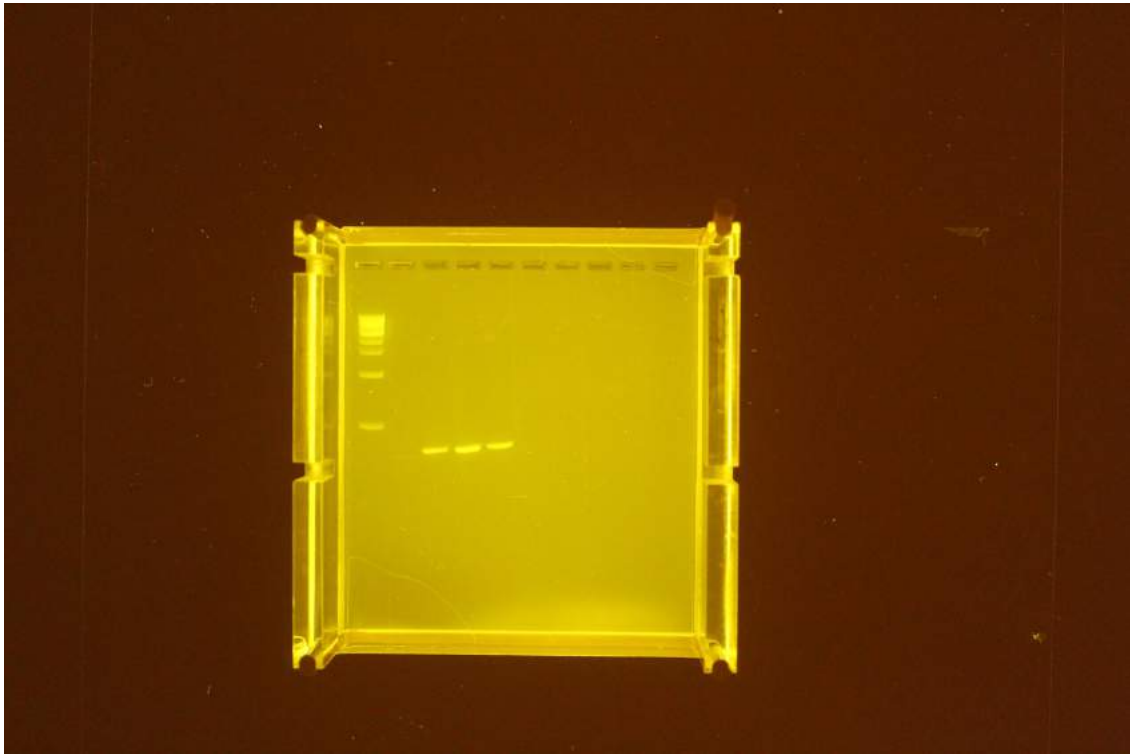
MONDAY, 31/07/2023

- Miniprep plasmids from transformation Week July 24 + Conc. Determination
 - **Result:**
 - The samples are already sent for sequencing

Table4			^
	A	B	
1	Sample	Conc. (ng/μL)	
2	1	198.09	
3	2	157.75	
4	3	106.13	

- Clone hydrophobin sequences into plasmid with linker
 - Miniprep and sequence linker plasmid (Aby)
 - PCR of linker plasmid (Vlad)
 - PCR of 4 hydrophobin sequences: HGFI, HGFI_mut_half, HGFI_mut_full, HFBII_mut (JeongHo)
 - Labelled as = HFBII_mut (1) // HGFI (2) // HGFI_mut_half (3) // HGFI_mut_full (4)
 - Performed [Taq polymerase PCR protocol](#)
 - Annealing temperature: 57°C for (1) // 46°C for (2), (3), and (4)
 - PCR overnight (the most right PCR machine: (1) in left lid // (2), (3), and (4) in right lid)
 - Running gel for looking at result of PCR amplification of hydrophobins [08.04.2023]
 - **Result:**
 - 2% agarose gel, **1kb** ladder (-> low mw dna ladder)
 - No band on 2nd lane (HFBII_mut)
 - bands appeared in HGFI (3rd lane), HGFI_mut_half (4th lane), and HGFI_mut_full (5th lane)

 Hydrophobin amplified



- PCR purification of amplified hydrophobin sequences + Conc. Determination (done in the next week)

TUESDAY, 01/08/2023

- Troubleshooting BsaI digest for Golden Gate
 - => Re-doing digestion of the PCR amplification of plasmid (pET29-sfGRP from Sisi) with longer incubation time (1hr instead of 30 mins).
 - Used 3 plasmid stock product amplified on 07.26.2023
 - Conc as 54.71 // 75.71 // 62.9 [ng/ul]
 - labelled as (1) // (2) // (3)
 - pET29_linker_insert_1, 2, and 3
 - labelled as (4) // (5) // (6)

Materials used:			^
	A	B	
1	Materials	Volume (µL)	
2	PCR product	45	
3	DpnI	1	
4	BsaI	1	
5	CutSmart Buffer	10	
6	MQ water	43	
7		100	

- PCR purification ([GeneJET PCR purification kit](#)) + Conc. measurment

Result			^
	A	B	
1	Sample	Conc. (ng/µL)	
2	1	26.94	
3	2	38.15	
4	3	38.4	
5	4	11.59	
6	5	35.3	
7	6	35.74	

WEDNESDAY, 02/08/2023

- Gibson assembly [primers](#) designed by Snapgene -> Ordered through Kiyo
- Troubleshooting
 - Kiyo's plasmid with 4 different restriction enzymes
 - BsaI // NruI HF // Nb.BbvCI // MfeI. See [troubleshooting BsaI page](#).

Annealed oligonucleotides to create SGGSGGS linker with 4 nt overhangs at 5' ends. Used annealing protocol from Kiyo. (mix 50 uL Fwd (1 nM) + 50 uL reverse (1 nM)).

THURSDAY, 03/08/2023

Phosphorylated duplex using T4 PNK:

Table1 ^

	A	B
1	Reagent	Volume
2	Annealed oligo duplex	20
3	10x T4 Buffer (with ATP)	5
4	Nuclease free water	24
5	T4 PNK	1

Concentration of oligonucleotides was too low from the stock. The phosphorylation and annealing protocol was changed to one adapted from the internet. Phosphorylation protocol: 37°C for 30 min, 65°C for 20 min.

[Link to protocol](#)

Concentration of annealed phosphorylated oligonucleotides:

Table3 ^

	A	B
1	Sample	Conc (ng/uL)
2	Blank	0.94
3	Annealed oligos (rep 1)	107.91
4	Annealed oligos (rep 2)	107.47

Ligation with digested vector backbone using T4 ligase for 2h at room temperature.

Troubleshooting Bsal digest for Golden Gate

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

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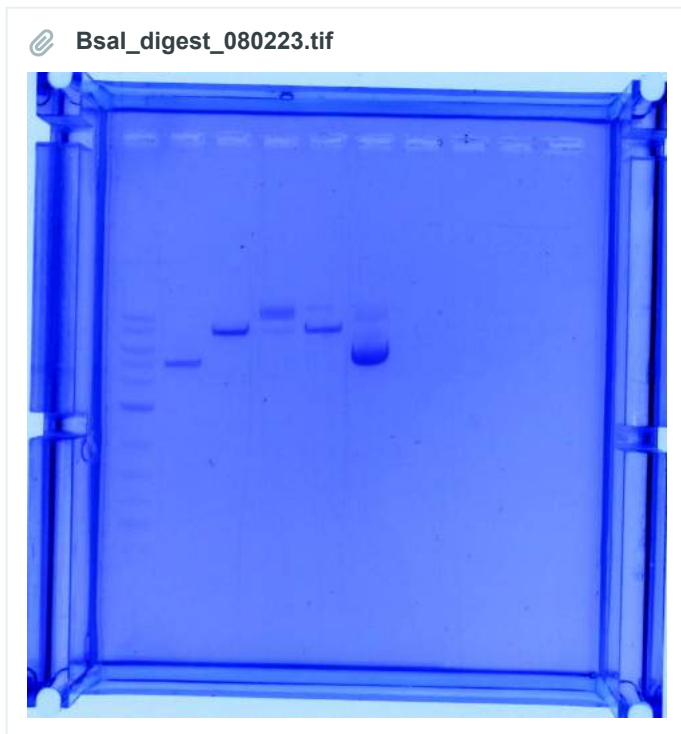
WEDNESDAY, 02/08/2023

Try Kiyo's plasmid with Bsal and BamHI sites.

1. Bsal only: 6 kb + 1 kb fragment
2. SapI only: unique cutter: 7 kb
3. Unique cutter
4. Unique cutter

Add 1 uL plasmid (200 ng)

Gel



Lane 1: NEB 1kb Plus Ladder

Lane 2: Plasmid + Bsal

Lane 3: Plasmid +

Lane 4: Plasmid +

Lane 5: Plasmid +

Lane 6: Supercoiled plasmid

Conclusion: Bsal enzyme is active, we can move on with repeating experiment for Golden Gate assembly

Summary of Week July 31 - Aug 4

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

Entry Created On: 04 Aug 2023 11:46:25 UTC

Entry Last Modified: 08 Aug 2023 14:10:09 UTC

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FRIDAY, 04/08/2023

This week we did:

- Miniprep on linker plasmid from last week:
 - Sequencing results showed the linker was not inserted. **Most likely cause:** Bsal digestion did not work
- Redigested amplified pET29 vector with Bsal, while also changing the incubation time (30 min to 1 hour)
- Amplified HGFI, HGFI_mut_half, HGFI_mut_full hydrophobin sequences

Week Aug 7 - 11

Project: iGEM KU Leuven 2023
Author: Vlad Batagui
Entry Created On: 07 Aug 2023 08:02:40 UTC
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MONDAY, 07/08/2023

- Planning for Monday:
- colony PCR of colonies from linker plasmid
 - purify hydrophobin PCR products (HGFI, HGFI_mut_half, HGFI_mut_full)
 - redo PCR for HFBII_mut

RESULTS

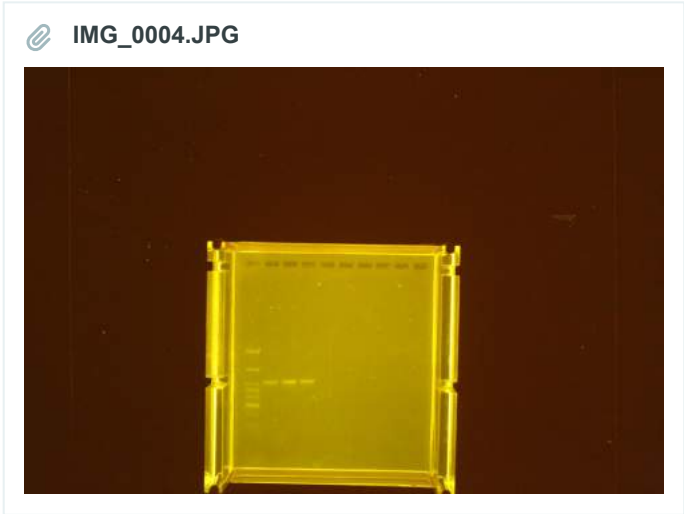
- **Purification hydrophobin PCR products (HGFI, HGFI_mut_half, HGFI_mut_full) (Brigitte)**

- [Protocol used for PCR purification](#)
- remark: included step with isopropanol (protocol A)
 - results concentration:

Table1 ^

	A	B
1	Sample	Conc. (ng/μL)
2	HGFI	41.7
3	HGFI_mut_half	55.8
4	HGFI_mut_full	41.51

- Running gel for looking at result of PCR amplification after purification
- **Result:**
 - 2% agarose gel, low MW ladder
- picture gel



- redo PCR for HFBII_mut
 - Protocol
 - reaction volume = 50 μ L
 - annealing temperature 57°C (<https://tmcaculator.neb.com/#!/main>)

Colony PCR with primers: pET29_link_vector_fwd pET29_link_vector_rev using Q5 master mix

TUESDAY, 08/08/2023

Miniprep colonies transformed with pET29-sfGFP-linker ligation product (Vector 1: 1-6, Vector 5: 6,7).
Colony PCR gave bands in colonies formed with vector 5 (6,7). Sent for sequencing V1: 5,6; V5: 6,7.

Miniprep plasmid concentrations ^		
	A	B
1	Sample	Conc. (ng/ μ L)
2	V5 6	62.4
3	V5 7	24.62
4		

Colony PCR

Digest colony PCR with Dpn1 and Bsal:

Materials used:		
	A	B
1	Materials	Volume (µL)
2	PCR product	48
3	DpnI	1
4	BsaI	1
5	CutSmart Buffer	10
6	MQ water	40
7		100



Purified PCR products using GeneJET kit.

PCR Colony purification concentr...		
	A	B
1	Sample	Conc. (ng/µL)
2	V5 6	57.5
3	V5 7	35.22
4		



Gibson PCR:

Q5 plasmid 2 protocol in iGEM folder in red thermocycler , with annealing at 60 C. Reverse primer

Gibson Vector PCR Mix		
	A	B
1	Reagent	Volume (uL)
2	Q5 2x Master Mix	25
3	Template DNA (10 ng/uL)	1
4	fwd primer (10 uM)	2.5
5	rev primer (10 uM)	2.5
6	Water	19

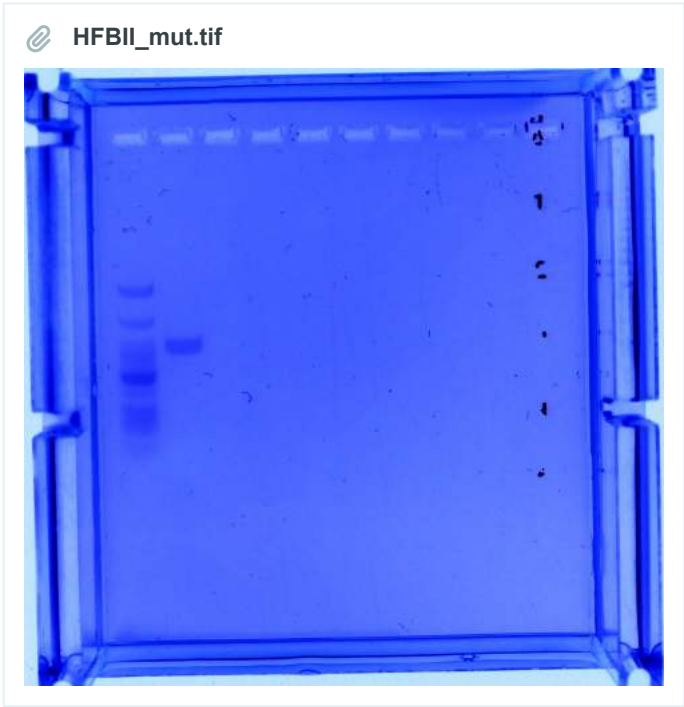


PCR could not be done due to primer for vector amplification missing.

Ran 2% gel of HFBII_mutated sequence amplified on 7/8/2023:

Gel image:

- Lane 1: Low MW Ladder (NEB)
- Lane 2: HFBII_mutated amplicon for Golden Gate cloning



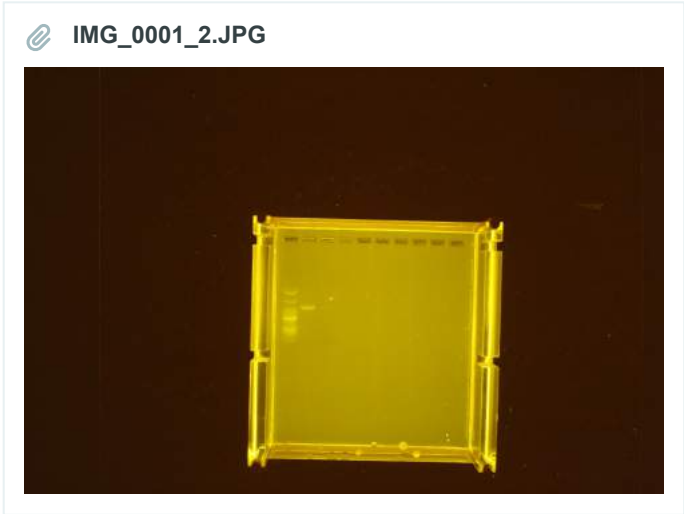
Purified PCR products using GeneJET kit (HFBII_mutated).

HFBII_mutated concentrations		
	A	B
1	Sample	Conc. (ng/μL)
2	HFBII_mutated	114.07
3		
4		

there are impurities in this sample.

Ran 2% gel of PCR product of HFBII_mutated sequence PCR purified

- Lane 1: Low MW Ladder (NEB)
- Lane 2: HFBII_mutated amplicon for Golden Gate cloning



WEDNESDAY, 09/08/2023

Digestion of PCR purified sequences: MBSP1, HFBI, HFBI mut, HFBII, HFBII mut, HGFI, HGFI mut half and HGFI mut full

- programme in thermocycler = iGEM folder → digestion 1h and inactivation for 20 min

Digestion of hydrophobin s... ^		
	A	B
1	Reagent	volume (uL)
2	Hydrophobin sequence	25
3	Bsal	1
4	Cutsmart Buffer	5
5	Water	19

PCR purified using GeneJET PCR purification kit using optional isopropanol step.

Sequencing results from August 9 confirmed that V5-colony 7 sample had the SGGSGGS linker correctly integrated into the pET29-sfGFP plasmid.

Transformed DH10-beta chemicompetent cells to generate stock of pET29-linker-sfGFP plasmid from V5-colony 7. Protocol used linked to this file.

Amplified V5-7 miniprep product with **pET29_link_vector_fwd pET29_link_vector_rev** using Q5 master mix, annealing temperature 58 °C.

THURSDAY, 10/08/2023

- Plan for the day:
- ligation of digested vector containing linker with digested hydrophobin sequences
 - transformation of ligation products
 - inoculate liquid culture to generate stock of pET29-linker-sfGFP plasmid

RESULTS

Ligation of digested vector containing linker with digested hydrophobin sequences

Vector and insert were added in a 1:5 molar ratio. Ligation mix was incubated for 2 hours at RT. Hi-T4 ligase activity was inactivated for 10 min at 65 °C.

Ligation hydrophobin sequence a... ^

	A	B
1	Reagent	Volume (uL)
2	V5 7 vector (50 ng)	2,03
3	T4 DNA Ligase Buffer (10X)	2
4	Hi-T4™ DNA Ligase	1
5	Insert	see calculation below
6	H2O	added to 20

The required insert mass was calculated with [NebCalculator](#).

Calculations for volume of hydrophobins added to ligation mix



	A	B	C	D	E
1	Hydrophobin sequence	Conc (ng/uL)	ng per reaction	uL added to reaction	
2	MBSP1	19,14	40,77	2,13	
3	HFB1	10,7	10,50	0,98	
4	HFB1 mut	11,18	10,50	0,94	
5	HFBII	10,23	13,36	1,31	
6	HFBII mut	9,48	13,36	1,41	
7	HGFI	14,52	14,73	1,01	
8	HGFI mut half	18,47	14,73	0,80	
9	HGFI mut full	12,5	14,73	1,18	

Transformation of E coli with ligated constructs (100 uL per cell aliquot)

Gibson assembly:

- started amplifying pET29 vector with primers for Gibson Assembly

Gibson PCR:

Q5 plasmid 2 protocol in iGEM folder in red thermocycler , with annealing at 60 C.

Gibson Vector PCR Mix1



	A	B
1	Reagent	Volume (uL)
2	Q5 2x Master Mix	25
3	Template DNA (10 ng/uL)	1
4	FWD primer (10 uM)	2.5
5	REV primer (10 uM)	2.5
6	Water	19

Stock generation of pET29-linker-sfGFP plasmid from V5-7 miniprep product from August 7.

Picked 3 colonies, inoculated for overnight culture.

FRIDAY, 11/08/2023

- Gibson PCR product purification**

On this page: Entry

Concentration of Gibson PCR Pro... ^

	A	B
1	Sample	Conc. (ng/μL)
2	1	80.5
3	2	48.5

- Sequence sent for 3 colonies of pET29-linker-sfGFP plasmid from V5-7

Concentration of pET29-linker-sfGFP plasmid fro... ^

	A	B	C
1	Sample	Conc. (ng/μL)	Sequencing number
2	1	26.55	231CZAD180
3	2	35.66	231CZAD181
4	3	24.67	231CZAD183

- Glycerol stock of the 3 colonies with pET29-linker-sfGFP plasmid
- Gibson PCR product digestion with DpnI

Concentration of Gibson PCR Pro... ^

	A	B
1	Sample	Conc. (ng/μL)
2	1	38.96
3	2	26.42

Week Aug 14 - 18

Project: iGEM KU Leuven 2023

Author: Brigitte Bais

Entry Created On: 14 Aug 2023 07:19:35 UTC

Entry Last Modified: 21 Aug 2023 09:28:36 UTC

Export Generated On: 10 Oct 2023 15:51:05 UTC

MONDAY, 14/08/2023

Started PCR amplification of hydrophobins for Gibson cloning ([protocol](#) used) for HFBII, HFBII_mut, HGFI, HGFI_mut_half HGFI_mut_full in reaction volumes of 50 μ l. Annealing temperatures [here](#).

TUESDAY, 15/08/2023

Gel from PCR products



- 1kb ladder (1) and low kb ladder (2)
- Showing bands for HFBII mut, HGFI, HGFI mut half (but not for HGFI mut full and HFBII)

PCR purifications of Gibson hydrophobins

Concentrations after purification			^
	A	B	
1	Sample	Conc. (ng/μL)	
2	HFBII mut	55.7	
3	HGFI	50.77	
4	HGFI mut half	52.76	
5	HGFI mut full	69.33	
6	HFBII	25.62	

WEDNESDAY, 16/08/2023

Gibson assembly reaction

Gibson assembly reaction mixture			^
	A	B	
1	Reagent	Volume (uL)	
2	Total amount of fragments	See below (X)	
3	Gibson Assembly Master Mix (2X)	10	
4	Deionized H2O	10-X	
5	Total Volume	Up to 20	
6			

The required insert mass was calculated with [NebCalculator](#). (1:3 ratio)

Calculations for volume of hydrophobins added to ligation mix



	A	B	C	D	E
1	Hydrophobin sequence	Conc (ng/uL)	ng per reaction	uL added to reaction	Total amount of fragments (vector + hydrophobin)
2	HFBII mut	55.7	15.98	0.29	2.86
3	HGFI	50.77	17.62	0.35	2.92
4	HGFI mut half	52.76	17.62	0.33	2.9
5	HGFI mut full	69.33	17.62	0.25	2.82
6					
7	Vector	38.96	100	2.57	
8					
9					

Transformed HGFI and HGFI_mut_half Gibson assembly products into Dh10Beta cells.

Innoculated overnight cultures of Yarrowia plasmids from INRAE and 8 Golden Gate constructs for E coli cloning

THURSDAY, 17/08/2023

Miniprep of Golden Gate constructs for E coli (8 constructs x 2 replicates) + 4 plasmids for Yarrowia (INRAE) + glycerol stocks

Issue with Yarrowia plasmids: plasmid DNA was sent instead of E coli on agar containing plasmid. Inoculation of LB media with DNA solution resulted in bacterial growth even in the presence of antibiotics -> potential contamination

FRIDAY, 18/08/2023

- Miniprep Gibson construct (HGFI 1/2 and HGFI mut half 2)

Table1



	A	B
1	Sample	Conc.
2	HGFI 1	20.16
3	HGFI 2	18.58
4	HGFI mut half 2	18.01

- Send for sequencing 8 hydrophobins constructs using Golden Gate + Gibsons HGFI and HGFI mut half

Table2		
	A	B
1	Sample	Label
2	HFBI 1	231CZAD151
3	HFBI mut 1	231CZAD152
4	HFBII 1	231CZAD153
5	HFBII mut 1	231CZAD154
6	HGFI 1	231CZAD155
7	HGFI mut half 1	231CZAD156
8	HGFI mut full 1	231CZAD157
9	MBSPI 1	231CZAD158
10	HGFI mut hm 2 (Gibson)	231CZAD165
11	HGFI 1 (Gibson)	231CZAD159
12	HGFI 2 (Gibson)	231CZAD160



Week Aug 21 - 25

Project: iGEM KU Leuven 2023
Author: Vlad Batagui
Entry Created On: 21 Aug 2023 08:39:29 UTC
Entry Last Modified: 10 Oct 2023 15:44:44 UTC
Export Generated On: 10 Oct 2023 15:51:05 UTC

MONDAY, 21/08/2023

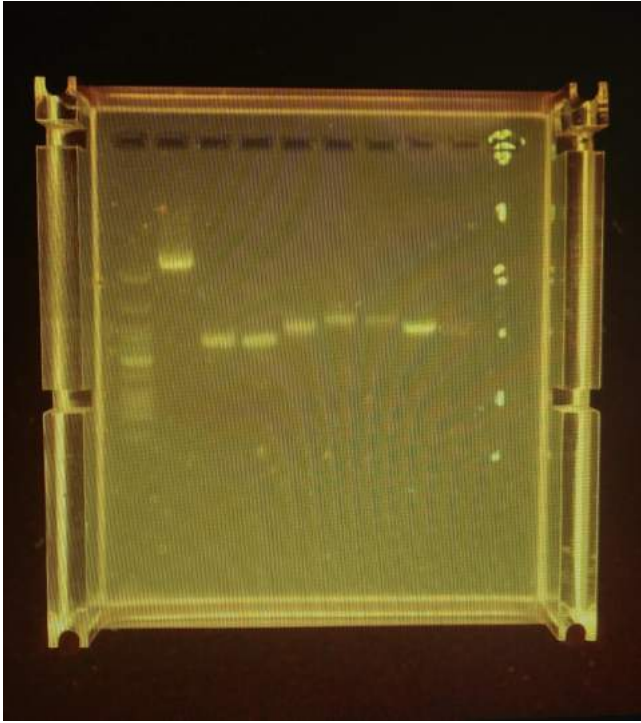
- Plan for the day:
- Send plasmid extracted from 8 new colonies for sequencing

Table1 ^

	A	B
1	Hydrophobin	Sequencing No.
2	HFBI	231CZAD182
3	HFBI mut	231CZAD144
4	HFBII	145
5	HFBII mut	146
6	HGFI	147
7	HGFI mut hm	148
8	HGFI mut fm	149
9	MBSPI	150
10		

- start new Gibson assembly for remaining hydrophobins (start with MBSP1, as it is the longest fragment)
 - To be purified on Tuesday
 - 1: MBSPI
 - 2: HFBI mut
 - 1. (blue): HFBI
- Run gel electrophoresis for the PCR of hydrophobins
 - in order from left to right: low mw ladder, MBSP1, HFBI mut, HFBI, HFBII mut, HGFI, HGFI mut hm, HGFI mut fm, HFBII

 20230821_165957.jpg



Transformed Dh10-beta cells with plasmids for Yarrowia (1. NDV-URA3, 2. P1-TEF, 3. TEF-8UAS, 4. Lip2 terminator)

TUESDAY, 22/08/2023

Inspected plates for transformation of Yarrowia plasmids, negative control cells grew on Amp plate, not on Kan plate. Testing Amp and Carb in liquid culture with Dh10-beta cells.

Re-transformed NDV-URA plasmid and plated on CAB plates instead of Amp

WEDNESDAY, 23/08/2023

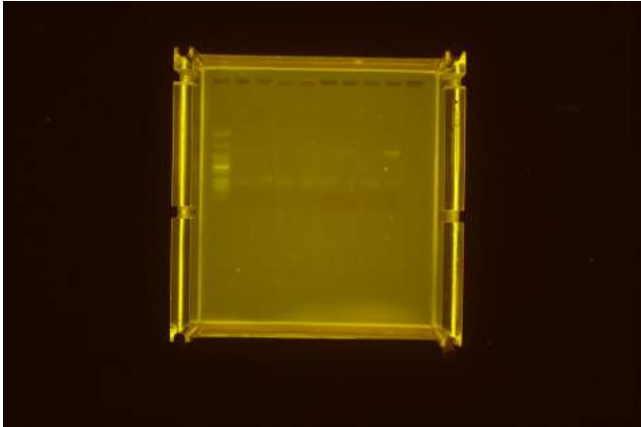
PCR amplified pET29-linker-sfGFP plasmid from miniprep on August 11 (Sample 2) using **pET29_link_vector_fwd** and **pET29_link_vector_rev** primers.

Digested with BsaI + DpnI (Zosia)

Colony PCR for Gibson assemblies of MBSP1 (8 colonies) and HFBI (6 colonies) with primers that target the hydrophobin inserts.

- Results: MBSP1 8th colony showed a strong band

IMG_0002_1.JPG



THURSDAY, 24/08/2023

PCR amplified pET29-linker-sfGFP plasmid from miniprep on August 11 (Sample 2) using **pET29_link_vector_fwd** and **pET29_link_vector_rev** primers.

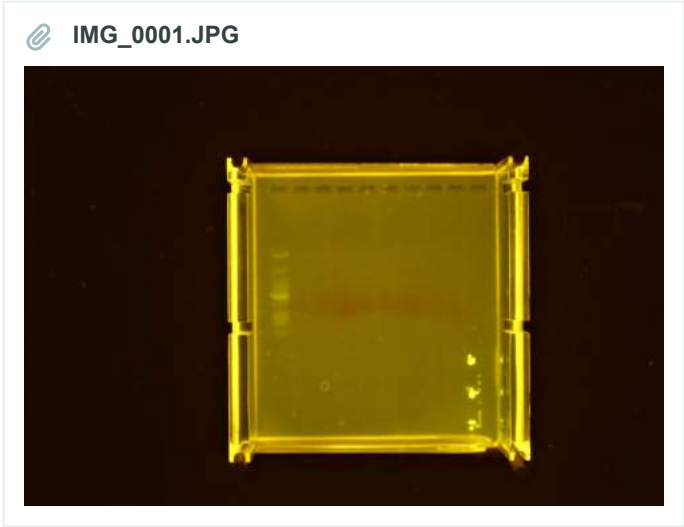
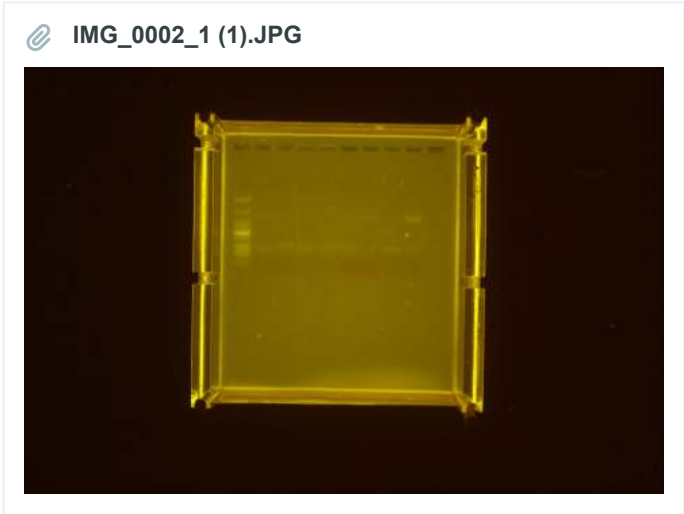
Digested with Bsal + Dpn1 (Zosia) - repeated again today bc yesterday nothing came up in the gel.

Digested it with Dpn1 and Bsal.

Tube label: pET29 plasmid 24.08.23



- Made glycerol stocks for
 1. NDV-URA3
 2. P1-TEF
 3. P1-8UAS-TEF
 4. Lip2 term
- PCR colony for HGFI and HGFI_mut_half
 - Result: HGFI: weak bands around the size of HGFI for 1st, 6th, and 8th (1st photo). HGFI_mut_half: no bands (2nd photo)



- Liquid culture for MBSP1 8th colony (3 liquid cultures were made)

FRIDAY, 25/08/2023

Miniprep and sending sequence for MBSP1 8th colony (1, 2 ,and 3)

Table2 ^

	A	B
1	Colony	Sequence No.
2	MBSP1 (8) 1	231CZAD124
3	MBSP1 (8) 2	125
4	MBSP1 (8) 3	126

Week Aug 28 - Sep 1

Project: iGEM KU Leuven 2023
Author: Brigitte Bais
Entry Created On: 29 Aug 2023 09:37:50 UTC
Entry Last Modified: 09 Oct 2023 21:56:20 UTC
Export Generated On: 10 Oct 2023 15:51:05 UTC

TUESDAY, 29/08/2023

Gibson ligation reaction for HFBI, HFBI mut, HFBII, HFBII mut
[Protocol used](#)
Per reaction 75 ng of vector (Gibson I). Reactions were performed in a 1:3 insert:vector ratio.

Calculations for µl of hydrophobins sequence per reaction					^
	A	B	C	D	
1	Fragment	ng	conc (ng/µl)	µl per reaction	
2	HFBI	9.450	45.85	0.21	
3	HFBI mut	9.450	42.17	0.22	
4	HFBII	12.01	25.62	0.46	
5	HFBII mut	12.01	55.9	0.21	

Transformed HFBI, HFBI mut, HFBII and HFBII mut in DH5alpha competent cells.

Resuspended Yarrowia gBlocks we received last week.

Protocol used:

 image.png

Instructions

Dried contents may appear as either a translucent film or a white powder. This variance does not affect the quality of the gBlocks® Gene Fragments.

Resuspending your gBlocks® Gene Fragments

1. Prior to opening, centrifuge the tube at a minimum of 3000 x g to ensure that the material is at the bottom of the tube.
2. Add TE to reach a final concentration of 10ng/μL.
3. Vortex briefly.
4. Incubate at 50°C for 20 minutes.
5. Briefly vortex and centrifuge.

Amplifying your gBlocks® Gene Fragments

- *For gBlocks® Gene Fragments ≤1kb, amplification can be performed using a high fidelity polymerase. To avoid sequence mutations due to amplification errors limit cycles to 12 or fewer.
- *For gBlocks® Gene Fragments >1kb, we do not recommend amplification.

For additional information please see

www.idtdna.com/gblockssupport or contact genes@idtdna.com

Amplification not performed

Inoculated liquid cultures for HGFI section 1,6, and 8 and MBSP1

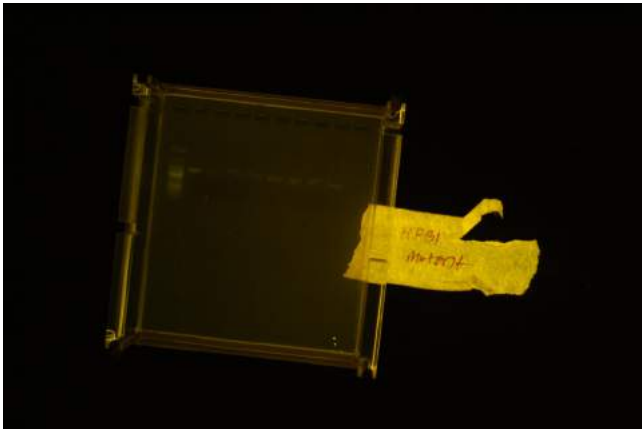
WEDNESDAY, 30/08/2023

Picked colonies for HFBI, HFBI mut, HFBII and HFBII mut and performed colony PCR.

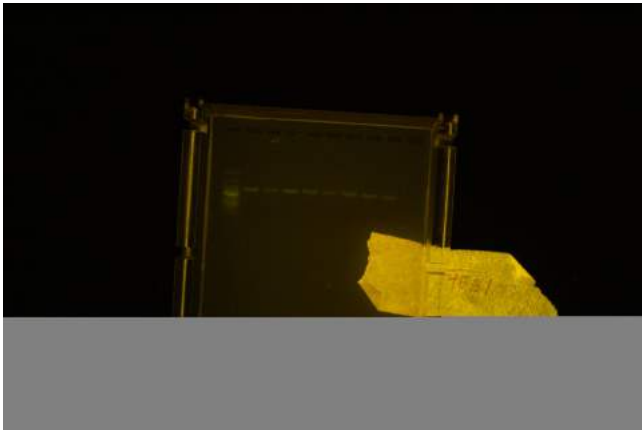
Results:

- Bands for 8 colonies of HFBI
- Bands for X colonies for HFBI mutant (1, 5,6,7,8)
- Unsure for HFBII, redoing gel on thursday
- Bands for HGFBII mutant (2,3)

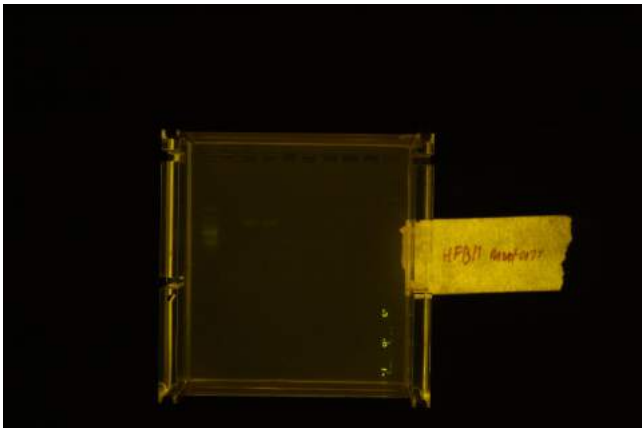
 HFBI mut.JPG

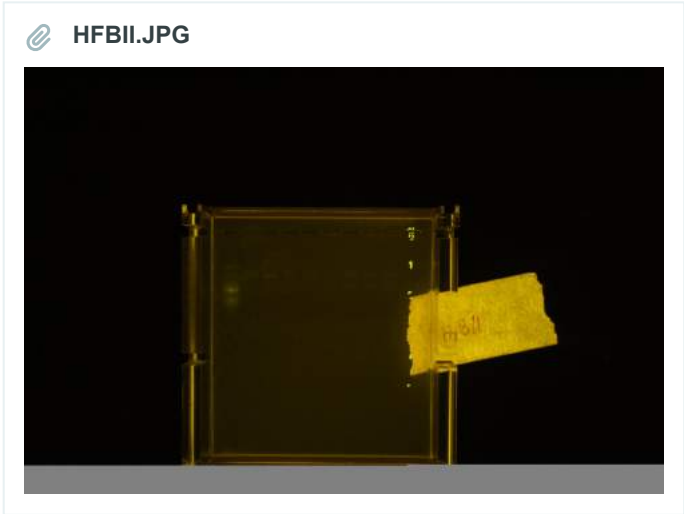


 HFBI.JPG



 HFBI mut.JPG





Inoculated cultures for:
HFBI --> 1,3,6
HFBI mutant --> 1,6,8
HFBII --> 1,2,3
HFBII mutant --> 2,3

THURSDAY, 31/08/2023

Not all liquid cultures grew.
HFBI colony 1,3,6 grew. Plan to send for sequencing.
HFBI mut 1,6,8 --> no growth
HFBII 1,2,3 --> grew but after redoing gel we saw no insert
HFBII mut 2,3 --> no growth

miniprep HFBI 1,3,6

Table1		
	A	B
1	Sequence	Conc. (ng/uL)
2	HFBI 1	10.72
3	HFBI 3	38.49
4	HFBI 6	10.51

For HFB1 mut picked 3,4,5 for inoculation of liquid culture
For HFBII there were colonies but the colony pcr showed empty bands so either the plasmids are empty or the colony PCR did not work. inoculated 3 more cultures with colonies 9, 10, 11. will perform colony pcr on those tomorrow.

no minipreps were send for sequencing today because there might be something wrong with the primers.

action item tomorrow:
miniprep HFBI mut 3,4,5
miniprep HFBII 9,10,11.
colony PCR HFBII 9,10,11.

FRIDAY, 01/09/2023

- Liquid culture shows that only HFBI mut 3,4 and HFBII 9,10 have clear indication of GFP
- Miniprep HFBI mut 3,4 and HFBII 9, 10
- Sequences were sent for HFBI 1,6,8 and HFBI mut 3,4
- HFBII 9,10 showed very faint bands on colony PCR around 300 bp. (repicked colony this morning after inoculating the culture yesterday so could be due to little template). HFBII 11 showed large insert and no GFP signal.

Table2

	A	B	C
1	Hydrophobin + colony number	Conc. (ng/uL)	Sequencing No.
2	HFBI 1	10.72	231CZAD103
3	HFBI 6	38.49	231CZAD102
4	HFBI 8	16.81	231CZAD101
5	HFBI mut 3	40.78	231CZAD100
6	HFBI mut 4	36.18	231CZAD099
7	HFBII 9	32.9	sequence on monday
8	HFBII 10	24.64	sequence on monday



action items next week
Clone HFBII mut, HGF1, HGF1 mut half, HGF1 mut full, start ligating on monday
HFBII 9,10 --> sequence on monday

Week Sep 4 - 8

Project: iGEM KU Leuven 2023
Author: Brigitte Bais
Entry Created On: 03 Sep 2023 17:26:33 UTC
Entry Last Modified: 10 Oct 2023 15:49:05 UTC
Export Generated On: 10 Oct 2023 15:51:05 UTC

MONDAY, 04/09/2023

Gibson ligation reaction for HFBII mut, HGF1, HGF1 mut half, HGF1 mut full. [Protocol used](#)
Per reaction 75 ng of vector (Gibson II). Reactions were performed in a 1:3 insert:vector ratio.
Required insert DNA mass per reaction was calculated using <https://nebiocalculator.neb.com/#!/ligation>

Calculations for hydrophobin sequences				
	A	B	C	D
1	Fragment	required insert DNA mass per reaction (ng)	conc (ng/μl)	μl per reaction
2	HFBII mut	12.03	55.9	0.22
3	HGF1	13.25	50.76	0.26
4	HGF1 mut half	13.25	69.33	0.19
5	HGF mut full	13.25	56.05	0.27

incubated vector + HiFi DNA assembly Master Mix + H2O at 50 degrees for 5 min followed by incubation with the inserts for remaining 10 minutes
transformed in DH5alpha competent cells

TUESDAY, 05/09/2023

- Colony PCR of HFBII mut, HGFI, HGFI mut half, HGFI mut full
- Send sequence (MBSPI is checked for the GFP sequence sing REV primer)
 - Result: no HFBII sequences were inserted. MBSPI (8) 1 and 2 showed complete sGFP sequence

Table2 ^

	A	B
1	Hyrophobins	Sequence No.
2	HFBII 9 FWD	231CZAD098
3	HFBII 10 FWD	231CZAD094
4	MBSPI (8) 1 REV	231CZAD095
5	MBSPI (8) 2 REV	231CZAD096

- Gel electrophoresis of PCR colony product of HFBII mut, HGFI, HGFI mut half, HGFI mut full
 - Result: bands for HFBII (1-8), HGFI (1-6), HGFI mut half (6), HGFI mut full (0)
- Liquid culture made: HFBII mut (1-4), HGFI (1-4), HGFI mut half (6)

WEDNESDAY, 06/09/2023

- Miniprepping HFBII mut (1,4), HGFI (1,2,3,4), HGFI mut half (6)

Table1 ^

	A	B
1	Hydrophobins	Sequence No.
2	HFBII mut 1 FWD	231CZAD083
3	HFBII mut 4 FWD	231CZAD084
4	HGFI 1 FWD	231CZAD085
5	HGFI 2 FWD	231CZAD086
6	HGFI 3 FWD	231CZAD087
7	HGFI 4 FWD	231CZAD088
8	HGFI mut half 6 FWD	231CZAD089

- Re-doing colony PCR for HGFI mut full (9-16)
- Gel electrophoresis for colony PCR HGFI mut full (gel electrophoresis)
 - Result: PCR failed

Week Sep 11 - 15

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

Entry Created On: 12 Sep 2023 12:40:32 UTC

Entry Last Modified: 04 Oct 2023 18:37:21 UTC

Export Generated On: 10 Oct 2023 15:51:04 UTC

TUESDAY, 12/09/2023

Plan for protein production. Transform BL21 cells with 3 constructs that were confirmed.

Day 1: Transformation and plating.

Day 2: Growth of overnight cultures

Day 3: Induction of expression and pelleting of cells

WEDNESDAY, 13/09/2023

Transformed BL21 cells with HFBI, HFBI_mut, MBSP1 plasmids (2 ul from each plasmid, between 50-75 ng per transformation)

Plated on Can

THURSDAY, 14/09/2023

Picked 2 colonies from each transformant and

FRIDAY, 15/09/2023

Started exponential cultures in the morning, until OD=0.2

Made glycerol stocks from overnight cultures (2x colonies per protein: HFBI, HFBI_mutant, MBSP1)

Week Sep 18 - 22

Project: iGEM KU Leuven 2023

Author: Brigitte Bais

Entry Created On: 19 Sep 2023 08:59:21 UTC

Entry Last Modified: 07 Oct 2023 13:24:47 UTC

Export Generated On: 10 Oct 2023 15:51:04 UTC

WEDNESDAY, 20/09/2023

Proteins were purified according to the protocol attached, with the exception of shaking the protein sample in resin for 1 hour at 37C.

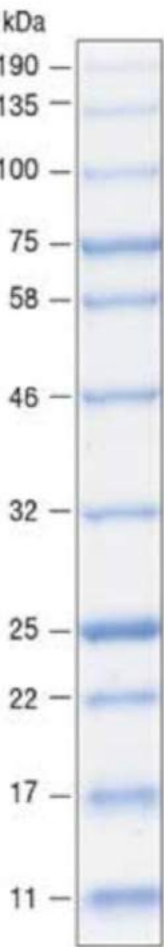
Results

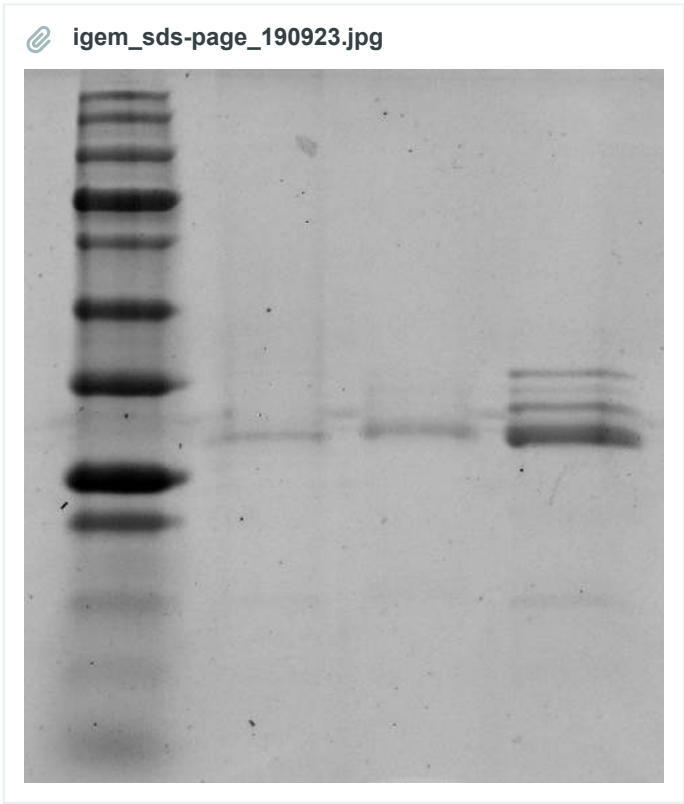
From left to right: MBSP1, HFB1 and HFB1 mut.

Ladder used: Blue Protein Standard Broad Range P7706S (NEB) (see attachment for sizes)

 NEB Protein Standard.jpg

Blue Protein Standard Broad Range
P7706S (New England Biolabs)





HFB11 minipreps were sent for sequencing

Table1 ^

	A	B
1	sample	sequencing number
2	HFBII (#1)	231CZAD029
3	HFBII (#2)	231CZAD030

Results: No insert for both minipreps.